Which AI claims in engine performance survive testing?

A measurement rig — and what it found, including about our own numbers

Start from the skeptical position, because the data supports it

This is not a pitch. It is an audit, and most of what was audited failed.

What we set out to test

- Does AI beat a *tuned* classical gas-path stack?
- Same data, same metrics, same tuning budget
- Hypotheses frozen before the runs
- Ground truth known, so a diagnosis is scored against the answer

What came back

- A published result's headline claim: noise
- Two of *our own* headlines: cut by 3–4×
- The task AI is most sold for: dead heat
- "Physics-informed AI": refuted twice

The part that survived points at **instrumentation**, not software.

The rig, in one slide

Everything from public certification data — TCDS and ICAO EEDB. No proprietary inputs.

- Thermodynamic model of a CFM56-7B26, calibrated to certification test-bed data
- 100-engine synthetic fleet run to failure, with **true** component condition recorded per flight
- Two tuned practitioners: classical Kalman gas-path analysis, and a matched AI suite
- Airline-format data: two snapshots per flight, cockpit sensors only

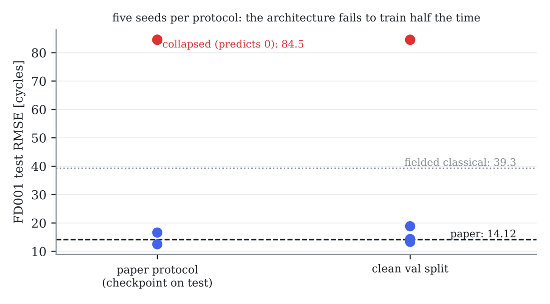
Why ground truth is the whole point

On real data you can only compare one estimate against another. Here we can ask the question that matters: *was the diagnosis right?*

What failed

A peer-reviewed deep-learning result, replicated exactly

Springer 2026: a bidirectional LSTM, "best of fifteen architectures" for remaining life.



It replicates. 14.56 cycles against a reported 14.12. Not a straw man.

But: the 15-way ranking rests on gaps of 0.12–0.60 cycles. Seed-to-seed spread alone is 2.94. The ranking is noise — one run per model.

And: the published architecture fails to train on 5 of 10 runs, predicting zero. Cause is in the paper's own tables.

The evaluation question to ask anyone, including us

Before you look at the headline

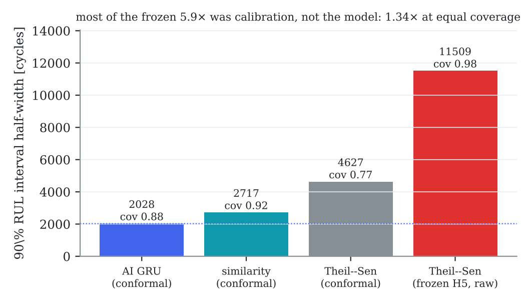
“What is the seed spread?”

- A ranking inside the run-to-run noise is not a result
- A model that silently fails half the time is not a product
- One run per architecture cannot support a ranking of fifteen

This alone is worth having when the next vendor deck arrives.

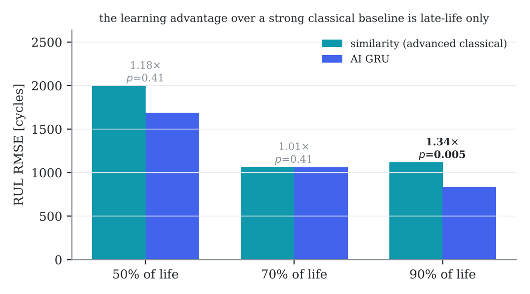
We applied it to our own claims, and both shrank

“6× tighter uncertainty”



Our interval was statistically calibrated; theirs was not. Calibrate both — free to the classical side — and it is 1.34×.

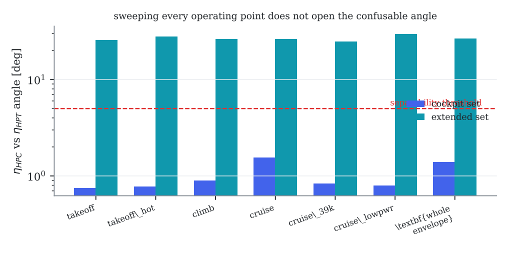
“2.3–4.4× better remaining life”



True only against the linear extrapolation fielded today. Against a competent classical method: 1.34× late in life, a tie at 70 %.

Where AI lost outright — and why no algorithm fixes it

Isolating confusable gas-path faults: 31 % vs 31 %. Identical.



Three cockpit measurements, ten health parameters, two fault signatures 1.3° apart. Confirmed four independent ways:

- halve every sensor noise → still 0.31–0.38
- add a “virtual sensor” → exactly zero
- sweep the whole operating envelope → 1.38°
- every learned method → no movement

No algorithm removes a rank deficiency.

And most prognostic error is not a method gap

87 % of mid-life remaining-life uncertainty is irreducible

Two engines in *identical measured condition* genuinely fail hundreds of cycles apart.
We can measure this because we know the true condition.

- Late in life a 4× reducible gap opens — **that** is where a better prognostic pays
- Early in life, a better algorithm is chasing the future, not a deficiency

A budget question answered with a number instead of an opinion.

What survived

An instrumentation design tool — and it is not AI

Classical estimation theory through the engine deck, checked against true condition.

Accumulate Fisher information over an engine's actual flown conditions $\Rightarrow$ a per-direction bound on what *any* diagnosis can know.

Instrumentation decision	Measured effect
Add station probes (P25/T25/PS3/T3)	HPC efficiency $45\times$ better recovered
Same probes, confusable isolation	$0.15 \rightarrow 0.92$
Halve noise on existing cockpit probes	no effect
"Virtual sensor" or better algorithm	no effect

It prices the hardware decision before the hardware is committed.

Honest caveat: the ranking claim behind this tool ($\rho = 0.70$ against true error) is real signal but does not clear its own physics-free null (0.24 median, $p = 0.085$) — the *acquisition* numbers above are separate measurements

Why this one is ours to act on

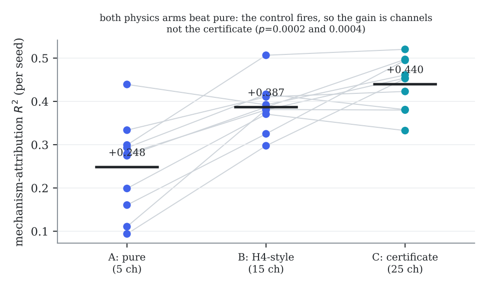
Product-strategy consequence

Any EHM service built on **cockpit-only data** sits under a hard informational ceiling that no software roadmap removes.

- The way through is **instrumentation** — which an OEM controls and a customer does not
- The analysis runs on **our own decks**, where it is stronger than on the public model used here
- It needs **no AI and no new data collection**

The one AI result that held up

And it is narrower and more useful than the usual claim.



Injecting physics into a learner:

- hurts remaining-life prediction (refuted twice)
- **helps** module attribution, by a large significant margin

Run with a **control arm**, so the gain cannot be attributed to simply adding more inputs.

Remaining life is already carried by the EGT-margin signal the model sees. *Which module* is exactly what a physical state estimate encodes.

What this is, and what it is not

Not

- A product or a validated operational tool
- Evidence about absolute numbers — the fleet is synthetic and the model is calibrated to public data, not our test cell
- A certified conclusion. These are statistical predictions

Is

- A reusable method for deciding which performance-AI claims to fund and which to reject
- Hypotheses frozen before runs; equal tuning budgets both sides
- Negatives reported with the same weight as positives
- Every figure regenerated from data, never transcribed

Proposed next step

Re-run the instrumentation analysis on our own engine decks and our own candidate sensor suites.

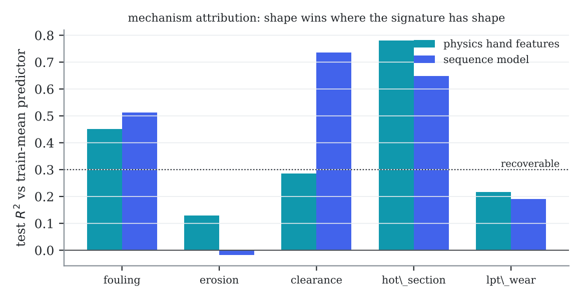
No AI. No new data collection. Answers a question we already pay to answer by other means.

Backup — the discipline that makes the answers usable

- **Pre-registration**: every hypothesis and threshold frozen in a tagged document before the run; refutations recorded, never renegotiated
- **Symmetric budgets**: both method families get the same number of tuning trials. Untuned comparisons inverted our own early scoreboard
- **Gates allowed to fail**: the dataset difficulty gate failed twice and the data was fixed, not the threshold
- **Built-in fraud checks**: give a method a case where physics says the information cannot exist, and confirm it finds nothing
- **Auto-generated figures**: no number reaches a slide without coming from the pipeline

Backup — the wall is a snapshot property

Three of five degradation mechanisms are attributable from the trajectory shape.



Including the hot-section mechanism that drives the confusable pair ($R^2 = 0.78$).

The rule we measured

Trajectory shape separates **what has shape**.

A wash sawtooth and a break-in knee are readable. A featureless linear ramp is not, however long you watch — which is also why a drifting sensor cannot be told from a real fault.